

EIGHT SESSION PACKET

Nuclear energy, taught in order

USA-first short path. Print one sheet per session or the whole packet. Kids wording lives on the website Adult/Kids toggle. This packet is the adult classroom version.

Numbers frozen September 2026. 94 U.S. reactors, about 18 percent of U.S. electricity, world share about 9 percent. Palisades loaded fuel 30 August 2026 and was still 0 percent power on 2 September 2026.

SESSION 1 OF 8

What nuclear energy is

Nuclear energy is heat from changes in an atomic nucleus. In a commercial plant that heat boils water, steam spins a turbine, and the turbine spins a generator. A coal plant does the last steps the same way. The difference is the heat source.

Fission splits a heavy nucleus — usually uranium-235. A neutron hits, the nucleus splits, heat and more neutrons come out. If those neutrons hit more uranium, you get a chain reaction. In a bomb the chain is built to run away. In a reactor, control rods catch neutrons on purpose so the chain stays steady.

Fusion joins light nuclei, the way the Sun does. U.S. plants you can visit today use fission. Fusion is still a lab and capital story.

Capacity is the size of the engine. Generation is how much it actually ran. Nuclear is a small slice of U.S. capacity and a much larger slice of generation because the engines run almost all year.

What we take from this. Nuclear power is a way to make a lot of heat from a small amount of fuel. The same science can make a weapon or a plant. The difference is control.

The other honest view. Calling nuclear “carbon-free” is sloppy. Lifecycle emissions are very low, not zero.

Try this. Knock over one domino, then two. Put your hand in the line. That is a control rod.

SESSION 2 OF 8

From discovery to the grid

The scientific path is short: X-rays and radioactivity in the 1890s, the neutron in 1932, fission in 1938–39. Szilárd saw the chain. Fermi’s team brought Chicago Pile-1 to a controlled chain reaction on 2 December 1942 under Stagg Field.

The first big use was a weapon. The Manhattan Project built Oak Ridge, Hanford, and Los Alamos. Trinity was July 1945. Hiroshima and Nagasaki followed in August. That is why “nuclear” still means “bomb” in public memory before it means “electricity.”

After the war, Congress moved the atom to a civilian commission (1946) and then allowed private plants (1954). EBR-I in Idaho lit four bulbs on 20 December 1951. Obninsk put nuclear electricity on a grid in 1954. Calder Hall (UK, 1956) was first at industrial scale. Shippingport (Pennsylvania, 1957) is the usual U.S. peaceful-commercial marker. The Navy’s Nautilus reactor is why most U.S. plants are light-water machines.

Eisenhower’s 1953 Atoms for Peace speech and the IAEA (1957) created the civilian-plus-safeguards frame. The Non-Proliferation Treaty (1970) is the later global promise: most countries skip weapons, inspectors can check plants, civilian use stays legal. The deal is incomplete. It still explains why a new “just electricity” program gets a diplomatic response a solar farm does not.

What we take from this. Civil nuclear is downstream of a weapons complex. Skipping that makes later fear and the NPT unintelligible.

Try this. Put three chips in order: Chicago pile, desert test, cities bombed.

SESSION 3 OF 8

Why America stopped building

The 1960s–70s were an order boom. PWRs and BWRs won. France stamped out look-alike plants after 1973. America let each utility design, litigate, and slip. Inflation, rising quality rules, slower demand growth, and cancelled orders hit before Three Mile Island.

TMI-2 (28 March 1979) was a partial core melt. Containment held. Public radiation doses were a sliver of background. Nobody in the public was killed by radiation. Cameras and confused official talk still taught a feeling: this can go wrong in our backyard. TMI turned a financial stall into a trust freeze.

Chernobyl (26 April 1986) was a different machine — an RBMK with a weak lid and graphite that could burn. INES 7. Early radiation deaths among firefighters and operators. Later thyroid cancers in children from iodine-131. Land emptied. America did not copy that design. America also did not start building again.

Fukushima Daiichi (11 March 2011) was a station blackout after earthquake and tsunami. Cores that were “off” still needed cooling. Three melted. UNSCEAR has not documented public radiation health effects the way Chernobyl’s firefighters were hurt. Evacuation and disrupted care did kill people. Cheap shale gas then made many U.S. merchant plants uneconomic. Vogtle 3 and 4 (2023–24) are the expensive exception that finished.

What we take from this. “We stopped building” is not “we stopped using.” The U.S. learned to run old plants at ~90% capacity factor and forgot how to pour new ones on time.

Try this. Two accident cards: Pennsylvania 1979 and Chernobyl 1986. Same word. Not the same day.

SESSION 4 OF 8

The grid today

Snapshot, September 2026. The United States has 94 operating commercial reactors at 54 plants in 28 states. About 97 GW net. Nuclear is roughly 8% of capacity and 18% of generation. 2025 capacity factor about 91%. Newest unit: Vogtle 4, Georgia, April 2024. Oldest running: Nine Mile Point 1, 1969. Large everyday-style U.S. reactors under construction: zero.

What is moving instead: restarts (Palisades in Michigan, Crane / former TMI-1 in Pennsylvania), uprates, and first advanced projects. TerraPower's Natrium plant in Wyoming received a construction permit in March 2026 and started construction in April. That is dirt, not megawatt-hours on a bill.

Illinois often gets about half its in-state electricity from nuclear — 11 reactors, the most of any state. Michigan's share slid toward ~19–22% after Palisades closed in 2022. Gas filled the hole. World: about 440 operable reactors, ~9% of electricity. China is connecting the new ones.

Nuclear's grid job is firm, clean, around-the-clock power from a few sites with small land use. It does not correlate with wind and sun. That is why data-center offtakers started shopping for it.

What we take from this. America still has the biggest nuclear-use story. It does not, right now, have the biggest nuclear-building story.

Try this. Open the map. Click your nearest pin. Is it running, waking, or gone?

SESSION 5 OF 8

Safety versus fear

Use two tracks. Track one is measured harm per unit energy. Track two is dread: invisible, delayed, tied to cancer, born as a weapon, remembered as three names.

A widely cited scoreboard (accidents plus air pollution, Our World in Data synthesis) puts brown coal near 33 deaths per TWh, coal near 25, oil near 18, gas near 2.8, hydro near 1.3, wind near 0.04, nuclear near 0.03, utility solar near 0.02. That is a model. It moves if you change Chernobyl's latent-cancer assumption. The robust point: routine fossil air pollution dominates energy's death toll.

TMI contained a partial melt. Chernobyl was a design-plus-test disaster with a large release. Fukushima was a cooling failure after a natural disaster; public radiation deaths are not established the way Chernobyl's early deaths are; evacuation harm is real.

Residual issues that are not "just perception": spent fuel still at plant sites, sabotage and wartime targeting (Zaporizhzhia), dual-use enrichment, cost overruns that become political risk, cooling water in a hotter climate, and who lives near mines and waste.

What we take from this. You can respect the fear and still read the record. The record says American plants have not killed the public with radiation. The record also says "never" is not a promise anyone honest makes.

Try this. Read TMI, Chernobyl, and Fukushima as three different days.

SESSION 6 OF 8

Michigan and the Great Lakes

National nuclear talk is abstract. Midwest nuclear is plants you can drive to.

Michigan operating: Donald C. Cook 1 & 2 at Bridgman on Lake Michigan (about 2,177 MW, AEP) and Fermi 2 near Monroe on Lake Erie (1,141 MW, DTE). Palisades in Covert Township closed 20 May 2022 and is the first U.S. attempt to leave decommissioning. Fuel loading began 30 August 2026. NRC status boards still showed 0% power on 2 September 2026. Restarting is not generating.

Fermi 1, on the Fermi 2 site, was a sodium-cooled fast breeder. On 5 October 1966 a loose plate blocked sodium flow. Two assemblies partially melted. Operators scrambled. No offsite release. John G. Fuller's book titled the public memory We Almost Lost Detroit. Teach the accident and the myth as two layers.

Illinois is the regional existence proof: about half of in-state generation, 11 reactors. Ohio still runs Perry and Davis-Besse — the latter carrying the 2002 vessel-head wastage scar. Iowa's Duane Arnold is in the same restart conversation, behind Palisades. Indiana has no commercial reactors and still rides the same wires.

What we take from this. If Palisades really returns, Michigan's mix and MISO reliability move more than any 2030s SMR slide deck.

Try this. Pin map: Cook, Fermi, Palisades, Big Rock. Colors: running, waking, gone.

SESSION 7 OF 8

LCOE versus firm clean watts

LCOE asks: if I build this plant and it runs as assumed, what is the average cost of each megawatt-hour? That is a useful fence-line number. It is not a grid plan.

Unsubsidized new-build ranges in Lazard-style 2025–26 reports typically put onshore wind and utility solar around \$40–100/MWh and new U.S. nuclear, Vogtle-anchored, around \$175–255/MWh. That chart is why people say new nuclear cannot compete. The chart under-weights firmness, capacity credit, multi-day weather droughts, and the fact that an existing plant is not Vogtle.

Vogtle 3 and 4: about 2.2 GW for a widely reported all-in bill on the order of \$32–35+ billion — roughly \$15,000–16,500 per kW. The only modern completed U.S. new-build receipt. Unit 4 was cheaper than Unit 3. “Nth-of-a-kind at \$6,000/kW” is a target. NuScale's cancelled Utah project printed a warning near \$20,000/kW on the way down.

Existing U.S. nuclear — fuel, people, maintenance — is a different object. Keep-running and credible restarts beat FOAK plants on time. Data-center PPAs are price discoveries for carbon-free power that does not vanish on a still winter night.

What we take from this. Old plants: cheap to keep if they are safe. New plants: expensive the first time, maybe less the fifth. Wind and solar: cheaper to begin, not a full copy of a plant that runs at 2 a.m. Anyone who tells you only one of those sentences is selling a side.

Try this. Two columns: keep the lakeside plant / replace with gas / replace with wind plus batteries.

SESSION 8 OF 8

What happens next

Three buckets. Bucket 1 — wake old plants. Palisades loaded fuel in August 2026. Crane (TMI-1) is a 2027-class target with a Microsoft PPA. Duane Arnold is behind both. Bucket 2 — first new kinds. Natrium is under construction in Wyoming. X-energy / Dow in Texas is in licensing. Light-water SMRs are a construction thesis not yet proven at U.S. commercial scale. Bucket 3 — talk. A political target to go from ~100 GW to 400 GW by 2050 is not an EIA forecast. America added only a few GW of new nuclear in the previous 25 years.

Fusion: NIF ignition in 2022 was a physics win, not a power plant. Company dates cluster in the late 2020s and early 2030s. Careful reviews often say pilot later, fleet after that. Keep three clocks — physics, machine duty-cycle, grid interconnection.

Waste: commercial spent fuel still sits at plants. Yucca Mountain remains the letter of the Nuclear Waste Policy Act and a political dead end. 2026 talk of consent-based “lifecycle campuses” is policy in motion, not a licensed repository. Finland is the Western benchmark for actually pouring a deep home.

What we take from this. The 2020s U.S. story is keep and restart. The 2030s might be build strange new ones. Do not mix those calendars.

Try this. Three clocks for fusion. None should say tomorrow unless a plant is selling power.

ENCYCLOPEDIA SHEET

Used fuel, casks, Yucca, and the NPT

After uranium splits for a few years the fuel is used. It is still hot and radioactive. It is not a mountain. For the electricity you got, the leftover pile is small.

Used fuel sits under water in a pool at the plant until it cools. Then it goes into dry casks — thick barrels on a pad. Those casks have a strong U.S. safety record. They were supposed to be a waiting room. The waiting room became the hotel.

The Nuclear Waste Policy Act sent the forever basement to Yucca Mountain in Nevada. Nevada refused. Congress stopped paying. Commercial spent fuel — on the order of 90,000 metric tons and rising — stays in the states that made the electricity. In 2026 the live political option is consent-based siting and talk of nuclear lifecycle campuses. That is policy in motion, not a licensed repository. Finland's Onkalo is the Western benchmark for actually pouring a deep home.

The same science that makes plant fuel can, if pushed much further, make weapons material. The Nuclear Non-Proliferation Treaty (in force 1970) is the global promise: most countries skip new bomb programs; inspectors can check plants; civilian use stays legal. Some countries never signed. Some cheated. Iran is the live compliance fight. The deal is incomplete and still explains why a new “just electricity” program gets a diplomatic response a solar farm does not.

Prairie Island in Minnesota makes the abstract local: two small operating PWRs beside the Prairie Island Indian Community, with dry casks on site for decades. Michigan's closed Big Rock Point still has casks inland from Lake Michigan.

What we take from this. The leftover fuel is a real unsolved American politics problem, not a bubbling cartoon barrel. The weapons promise is why nuclear electricity is never only an electricity story.

The other honest view. Reprocessing shrinks volume and does not erase the need for a repository. It also raises proliferation and cost arguments. Neither “leave it at the plant forever” nor “reprocess and the problem vanishes” is a finished answer.

HOW TO READ A HEADLINE**Four questions**

- Is it on? Making power today, being built, or only described?
- Who pays? Company, taxpayer, ratepayer, or a data-center contract?
- Compared with what? Coal smoke, wind plus batteries, or a new gas plant?
- Which accident? Pennsylvania 1979, Ukraine 1986, or Japan 2011 are not the same day.

Curiosity beats cheering and beats panicking.